Explainable AI Clinical Decision Support System

Clinical Prediction Report

Condition: Chronic Kidney Disease

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Patient Information

Field	Value
Patient ID	PAT003
Patient Name	Final Test Patient
Doctor / Clinician	Dr. Smith
Hospital / Clinic	Fortis Hospital
Disease	Chronic Kidney Disease

Clinical Parameters

Feature	Observed Value
Age	40
Blood Pressure	80
Specific Gravity	1.02
Albumin	0
Sugar	0
Red Blood Cells	0
Pus Cell	1
Pus Cell Clumps	0
Bacteria	0
Blood Glucose	100
Blood Urea	30
Serum Creatinine	1
Sodium	140
Potassium	4.5
Hemoglobin	15
Packed Cell Volume	45
White Blood Cell Count	8000

Feature	Observed Value
Red Blood Cell Count	5
Hypertension	Yes
Diabetes Mellitus	Yes
Coronary Artery Disease	Yes
Appetite	0
Pedal Edema	Yes
Anemia	Yes

Model Assessment

Field	Value
Condition	Chronic Kidney Disease
Prediction	Chronic Kidney Disease
Class	Chronic Kidney Disease
Probability	64.00%
Risk Level	Moderate
Clinical Recommendation	The prediction indicates some signs associated with chronic kidney disease. Consultation with a healthcare professional and further kidney function evaluation are recommended.

SHAP Evidence

Feature contribution to the model decision.

Feature	Contribution
Diabetes Mellitus	0.2491
Hypertension	0.0992
Hemoglobin	-0.0876
Packed Cell Volume	-0.0655
Serum Creatinine	-0.0584

LIME Evidence

Local explanation for this patient.

Feature / Rule	Weight
hemo > 0.81	-0.0987

Feature / Rule	Weight
-0.69 < dm <= 1.46	0.0777
-0.76 < htn <= 1.31	0.0582
-0.51 < sg <= 0.42	-0.0494
-0.36 < sc <= -0.31	-0.0493

SHAP-LIME Agreement

Consistency between the SHAP and LIME local explanation methods.

Field	Value
Agreement Score	84.50%
Feature Overlap	80.00%
Direction Agreement	100.00%
Ranking Agreement	75.00%
Overall Consistency	84.50%
Consistency Level	High
Shared Feature Count	4
Shared / Common Features	Dm, Htn, Hemo, Sc
SHAP Only	Pcv
LIME Only	Sg

Agreement measures consistency between SHAP and LIME explanations. It indicates explanation consistency, not clinical certainty.

Multi-Feature Counterfactual Explanation

Counterfactual explanations identify hypothetical alternative patient profiles in which one or more clinical features are changed to determine whether the model prediction can change.

Field	Value
Original Prediction	Chronic Kidney Disease
Original Probability	64.00%

Counterfactual #1

Field	Value
New Prediction	No Chronic Kidney Disease
New Probability	30.00%
Features Changed	1

Field	Value
Change Magnitude	1.00
Changed Clinical Features	Diabetes Mellitus: 1 → 0

Counterfactual #2

Field	Value
New Prediction	No Chronic Kidney Disease
New Probability	30.00%
Features Changed	2
Change Magnitude	6.00
Changed Clinical Features	Age: 40 → 35 Diabetes Mellitus: 1 → 0

Counterfactual #3

Field	Value
New Prediction	No Chronic Kidney Disease
New Probability	30.00%
Features Changed	2
Change Magnitude	6.00
Changed Clinical Features	Age: 40 → 45 Diabetes Mellitus: 1 → 0

Counterfactual #4

Field	Value
New Prediction	No Chronic Kidney Disease
New Probability	30.00%
Features Changed	2
Change Magnitude	11.00
Changed Clinical Features	Age: 40 → 30 Diabetes Mellitus: 1 → 0

Counterfactual #5

Field	Value
New Prediction	No Chronic Kidney Disease
New Probability	30.00%
Features Changed	2
Change Magnitude	21.00

Field	Value
Changed Clinical Features	Blood Pressure: 80 → 60 Diabetes Mellitus: 1 → 0

What does this mean?

Counterfactual explanations show hypothetical patient profiles in which one or more model input features are changed and the resulting prediction is evaluated. These results illustrate the model's decision boundary.

Important: Counterfactual scenarios are model-generated hypothetical examples. They must not be interpreted as clinical treatment targets, prescriptions, diagnoses, or instructions to change a patient's health measurements.

Clinical Interpretation Note

The prediction and explanation values in this report represent the output of the machine-learning system for the submitted patient data. SHAP and LIME provide model interpretability, while counterfactual analysis illustrates hypothetical changes that may cross the model's decision boundary.

Disclaimer: This report is intended for clinical decision support only. Model outputs, SHAP values, LIME explanations, agreement scores, and counterfactual scenarios must not be interpreted as definitive diagnoses, prescriptions, treatment instructions, or required clinical targets. Final clinical decisions must be made by qualified healthcare professionals using appropriate clinical assessment and additional evidence.

Generated by Explainable AI Clinical Decision Support System

Machine Learning • SHAP • LIME • SHAP-LIME Agreement • Counterfactual Analysis • Flask • React

Doctor Signature

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Dr. Smith

Patient Signature

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Final Test Patient